

# HUNDRED DAYS OF CODE

## DAY 17

### QUESTION:

Write a menu-driven C program using **switch-case** to perform digit-based operations on a positive integer. Define **one separate user-defined function for each menu option**, with *void return type* and *with a simple function call (without call by value/reference)*. Each function should read the required input, process it, and display the result. Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

- Generate the largest possible number using its digits
- Generate the smallest possible number using its digits.
- Swap its first and last digits.
- Check whether it is a palindrome number.

### 1. Algorithm

1. Start.
2. Display the menu:
  - 1. Generate the largest number using its digits.
  - 2. Generate the smallest number using its digits.
  - 3. Swap the first and last digits.
  - 4. Check whether the number is a palindrome.
  - 5. Exit.
3. Read the user's choice.
4. Use switch-case to select the corresponding function.
5. For option 1:
  - Read a positive integer.
  - Extract all digits.

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- Sort the digits in descending order.
  - Display the resulting largest number.
6. For option 2:
- Read a positive integer.
  - Extract all digits.
  - Sort the digits in ascending order.
  - Handle leading zero appropriately.
  - Display the resulting smallest number.
7. For option 3:
- Read a positive integer.
  - Extract the first and last digits.
  - Swap them.
  - Display the resulting number.
8. For option 4:
- Read a positive integer.
  - Reverse the number.
  - Compare the reversed number with the original.
  - Display whether it is a palindrome.
9. Repeat the menu until the user selects Exit.
10. Stop.

## 2. Pseudocode

BEGIN

REPEAT

DISPLAY menu

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READ choice

SWITCH(choice)

CASE 1:

CALL largest()

BREAK

CASE 2:

CALL smallest()

BREAK

CASE 3:

CALL swapFirstLast()

BREAK

CASE 4:

CALL palindrome()

BREAK

CASE 5:

DISPLAY "Program terminated"

STOP

DEFAULT:

DISPLAY "Invalid choice"

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END SWITCH

UNTIL choice = 5

END

**Function: largest()**

BEGIN largest

    READ number

    Extract all digits into an array

    Sort digits in descending order

    Display digits from left to right

END

**Function: smallest()**

BEGIN smallest

    READ number

    Extract all digits into an array

    Sort digits in ascending order

    If the first digit is 0:

        Find the first non-zero digit

        Place it at the beginning

    Display the digits

END

**Function: swapFirstLast()**

BEGIN swapFirstLast

    READ number

    Find first digit

    Find last digit

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Replace first digit with last digit

Replace last digit with first digit

Display resulting number

END

**Function: palindrome()**

BEGIN palindrome

    READ number

    Store original number

    Reverse the number

    IF original = reversed

        Display "Palindrome"

    ELSE

        Display "Not a palindrome"

END

### 3. C Program

```
#include <stdio.h>
```

```
/* Function prototypes */
```

```
void largest();
```

```
void smallest();
```

```
void swapFirstLast();
```

```
void palindrome();
```

```
int main()
```

```
{
```

```
    int choice;
```

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```
do
{
    printf("\n===== DIGIT OPERATIONS MENU =====\n");
    printf("1. Generate the largest possible number\n");
    printf("2. Generate the smallest possible number\n");
    printf("3. Swap first and last digits\n");
    printf("4. Check whether number is palindrome\n");
    printf("5. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            largest();
            break;

        case 2:
            smallest();
            break;

        case 3:
            swapFirstLast();
            break;

        case 4:
```

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```
    palindrome();  
    break;  
  
    case 5:  
        printf("Program terminated.\n");  
        break;  
  
    default:  
        printf("Invalid choice! Please try again.\n");  
    }  
  
} while (choice != 5);  
  
return 0;  
}  
  
/* Function to generate the largest possible number */  
void largest()  
{  
    long long n;  
    int digits[20], count = 0;  
    int i, j, temp;  
  
    printf("Enter a positive integer: ");  
    scanf("%lld", &n);
```

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```
while (n > 0)
{
    digits[count] = n % 10;
    n = n / 10;
    count++;
}

/* Sort digits in descending order */
for (i = 0; i < count - 1; i++)
{
    for (j = i + 1; j < count; j++)
    {
        if (digits[i] < digits[j])
        {
            temp = digits[i];
            digits[i] = digits[j];
            digits[j] = temp;
        }
    }
}

printf("Largest possible number: ");

for (i = 0; i < count; i++)
{
    printf("%d", digits[i]);
}
```

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```
printf("\n");
}

/* Function to generate the smallest possible number */
void smallest()
{
    long long n;
    int digits[20], count = 0;
    int i, j, temp, pos;

    printf("Enter a positive integer: ");
    scanf("%lld", &n);

    while (n > 0)
    {
        digits[count] = n % 10;
        n = n / 10;
        count++;
    }

    /* Sort digits in ascending order */
    for (i = 0; i < count - 1; i++)
    {
        for (j = i + 1; j < count; j++)
        {
```

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```
if (digits[i] > digits[j])
{
    temp = digits[i];
    digits[i] = digits[j];
    digits[j] = temp;
}
}
}

/*
If zero is the first digit, move the first
non-zero digit to the beginning.
*/
if (digits[0] == 0 && count > 1)
{
    pos = 1;

    while (pos < count && digits[pos] == 0)
    {
        pos++;
    }

    if (pos < count)
    {
        temp = digits[0];
        digits[0] = digits[pos];
        digits[pos] = temp;
    }
}
```

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```
    }  
}  
  
printf("Smallest possible number: ");  
  
for (i = 0; i < count; i++)  
{  
    printf("%d", digits[i]);  
}  
  
printf("\n");  
}  
  
/* Function to swap the first and last digits */  
void swapFirstLast()  
{  
    long long n, original, divisor = 1;  
    int first, last;  
    long long result;  
  
    printf("Enter a positive integer: ");  
    scanf("%lld", &n);  
  
    original = n;  
  
    /* Find the divisor for the first digit */
```

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```
while (n / divisor >= 10)
{
    divisor = divisor * 10;
}

first = original / divisor;
last = original % 10;

/* For a single-digit number, no change is needed */
if (divisor == 1)
{
    result = original;
}
else
{
    /*
        Remove first and last digits and swap them.
    */
    result = original;

    result = result - (long long)first * divisor;
    result = result - last;

    result = result + (long long)last * divisor;
    result = result + first;
}
```

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```
printf("Number after swapping first and last digits: %lld\n", result);  
}
```

```
/* Function to check palindrome */
```

```
void palindrome()
```

```
{
```

```
    long long n, original, reversed = 0, digit;
```

```
    printf("Enter a positive integer: ");
```

```
    scanf("%lld", &n);
```

```
    original = n;
```

```
    while (n > 0)
```

```
    {
```

```
        digit = n % 10;
```

```
        reversed = reversed * 10 + digit;
```

```
        n = n / 10;
```

```
    }
```

```
    if (original == reversed)
```

```
    {
```

```
        printf("%lld is a palindrome number.\n", original);
```

```
    }
```

```
    else
```

```
    {
```

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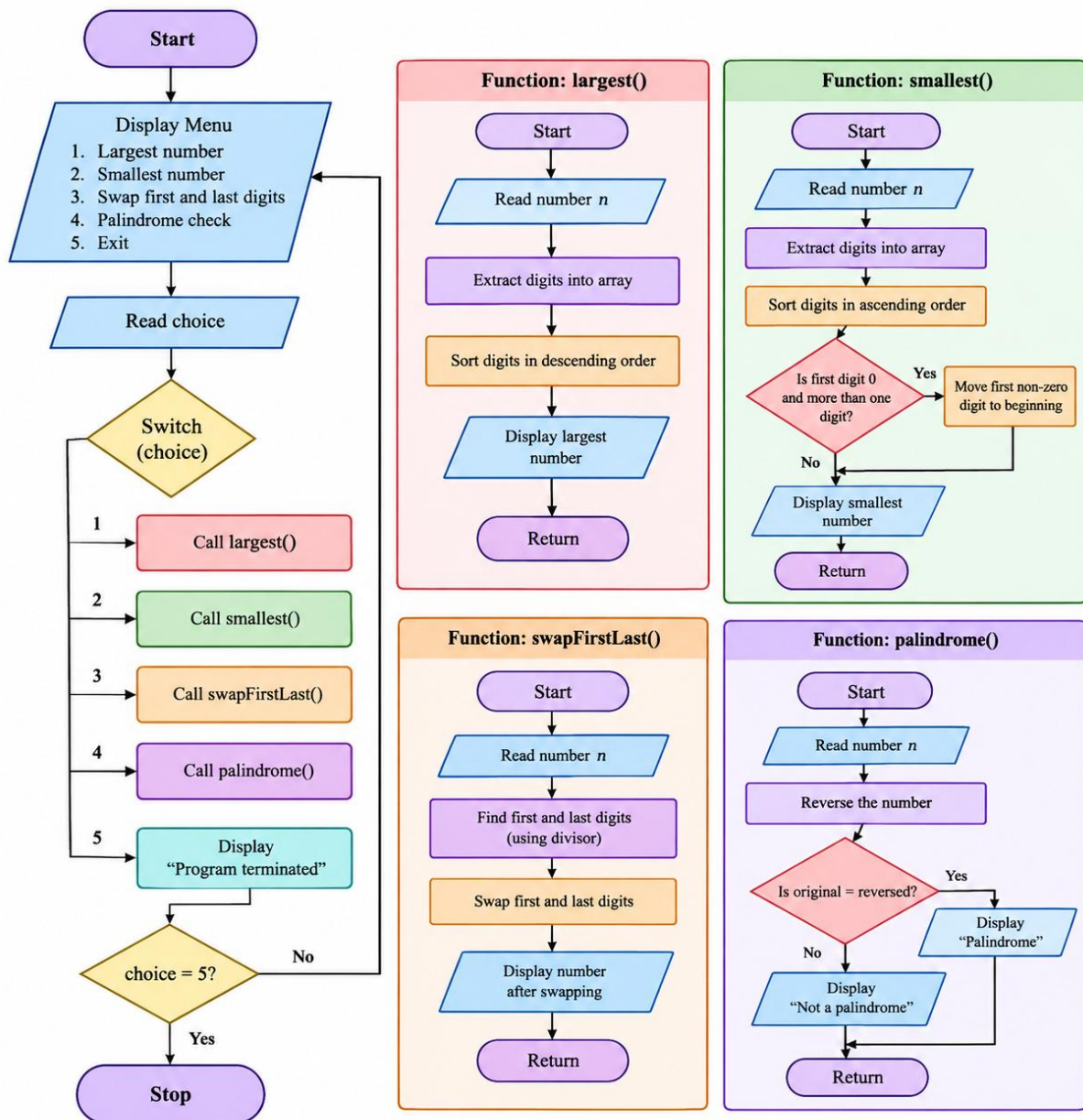
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```
printf("%lld is not a palindrome number.\n", original);
```

```
}
```

```
}
```

### 4. FLOWCHART :



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TEST CASE TABLE:

| TEST CASE | MENU CHOICS | INPUT | EXPECTED OUTPUT                 |
|-----------|-------------|-------|---------------------------------|
| 1         | 1           | 58321 | Largest possible number= 85321  |
| 2         | 2           | 40521 | Smallest possible number= 10245 |
| 3         | 3           | 9876  | Number after swapping=6879      |
| 4         | 4           | 1221  | 1221 is a palindrome            |
| 5         | 5           | --    | PROGRAM TERMINATED              |

COMPILATION AND EXECUTION:

```
(kali㉿kali)-[~/day17]
$ gcc palidrome.c -o palidrome

(kali㉿kali)-[~/day17]
$ ./palidrome

===== DIGIT OPERATIONS MENU =====
1. Generate the largest possible number
2. Generate the smallest possible number
3. Swap first and last digits
4. Check whether number is palindrome
5. Exit
Enter your choice: 1
Enter a positive integer: 58321
Largest possible number: 85321

===== DIGIT OPERATIONS MENU =====
1. Generate the largest possible number
2. Generate the smallest possible number
3. Swap first and last digits
4. Check whether number is palindrome
5. Exit
Enter your choice: 2
Enter a positive integer: 40521
Smallest possible number: 10245
```

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```
===== DIGIT OPERATIONS MENU =====
1. Generate the largest possible number
2. Generate the smallest possible number
3. Swap first and last digits
4. Check whether number is palindrome
5. Exit
Enter your choice: 3
Enter a positive integer: 9876
Number after swapping first and last digits: 6879

===== DIGIT OPERATIONS MENU =====
1. Generate the largest possible number
2. Generate the smallest possible number
3. Swap first and last digits
4. Check whether number is palindrome
5. Exit
Enter your choice: 4
Enter a positive integer: 1221
1221 is a palindrome number.

===== DIGIT OPERATIONS MENU =====
1. Generate the largest possible number
2. Generate the smallest possible number
3. Swap first and last digits
4. Check whether number is palindrome
5. Exit
Enter your choice: 5
Program terminated.

(kali㉿kali)-[~/day17]
$ █
```

.....